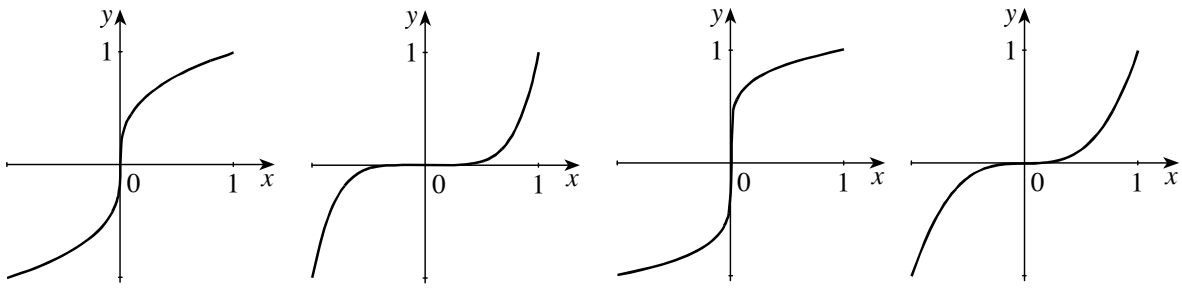
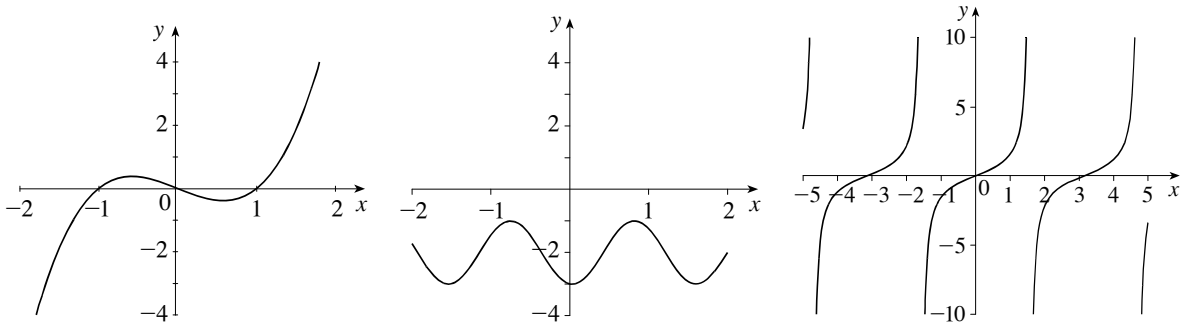
Warmup 1.2: Mathematical Models

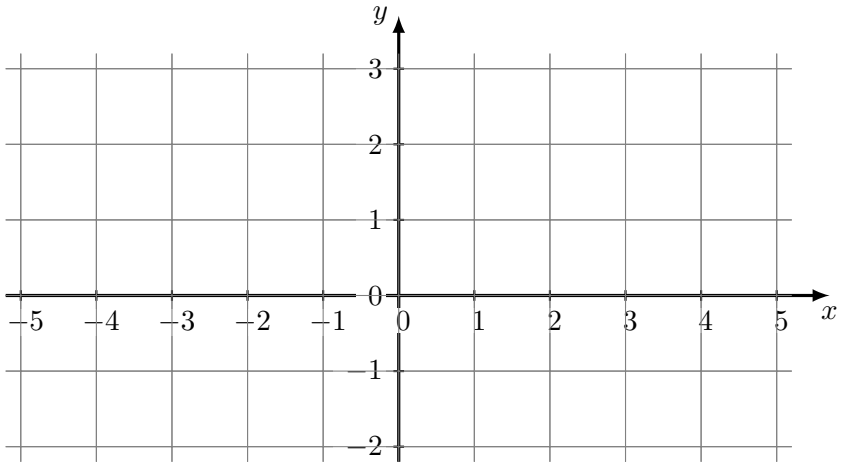
1. The four functions graphed below represent the functions $x^3, x^5, \sqrt[3]{x}, \sqrt[5]{x}$. But which is which?



2. Classify each function graphed below as a power function, root function, polynomial, rational function, algebraic function, or trigonometric function. Explain your reasoning.



3. Plot the rational Function $\boxed{f(x) = \dfrac{(x-1)}{(x-2)}}$



4. Find the domain of $\boxed{f(x) = \dfrac{1}{1 - \tan(x)}}$

5. Find an expression for a cubic function f if f(1) = 6, and f(-1) =f(0) =f(2) =0